CONFIGURATIONS OF LIMPOPO AND EASTERN CAPE BRANCHES

Step 1: Devices & Connections for Limpopo

- **HO Router (R-HO)**
  - **Has multiple WAN interfaces/subinterfaces (or serial links, depending on your setup).**
  - **Will connect directly to R-LIM (Limpopo router).**
- **Limpopo Router (R-LIM)**
  - **G0/0 = WAN link to HO (e.g. 10.22.241.2/30).**
  - **G0/1 = connected to local switch (Router-on-a-Stick).**
  - **Will provide DHCP & default gateway for Limpopo PCs.**
- **Limpopo Switch (SW-LIM)**
  - **Catalyst 2960 (layer 2).**
  - **Access ports for 10 PCs + 1 Printer.**
  - **One trunk port to R-LIM (for VLANs if needed).**
- **End Devices**
  - **10 PCs.**
  - **1 Printer (put in a Mgmt/Printer VLAN).**

🔹 **Step 2: Addressing Plan (Class A Package A)**

- **WAN Link HO ↔ LIM**
  - **R-HO (G0/2): 10.22.241.1/30**
  - **R-LIM (G0/0): 10.22.241.2/30**
- **Limpopo LAN (Single VLAN, +Printer)**
  - **VLAN 130 = 10.22.32.0/27 (covers 10 PCs + 1 Printer)**
  - **Gateway: 10.22.32.1 (R-LIM G0/1.130)**
  - **DHCP range: 10.22.32.10 – 10.22.32.30**

- **Printer static IP: 10.22.32.5**

🔹 **Step 3: Wiring**

- **R-HO G0/2 ↔ R-LIM G0/0 → Copper Cross-Over (if routers directly connected)**
- **R-LIM G0/1 ↔ SW-LIM Fa0/24 → Copper Straight-Through (set as trunk)**
- **SW-LIM Fa0/1–10 ↔ PCs → Copper Straight-Through**
- **SW-LIM Fa0/11 ↔ Printer → Copper Straight-Through**

Step 4: Configuration (CLI Snippets)

Limpopo Router (R-LIM)

```
! WAN link
interface g0/0
 ip address 10.22.241.2 255.255.255.252
 no shutdown

! Router-on-a-stick for LAN
interface g0/1
 no shutdown
!
interface g0/1.130
 encapsulation dot1q 130
 ip address 10.22.32.1 255.255.255.224

! DHCP for VLAN 130
ip dhcp pool LIMPOPO-POOL
 network 10.22.32.0 255.255.255.224
 default-router 10.22.32.1
 dns-server 10.22.0.130
```

```
 lease 2

! Default route to HO
ip route 0.0.0.0 0.0.0.0 10.22.241.1
```

Limpopo Switch (SW-LIM)

```
! Access ports for PCs
interface range fa0/1 - 10
 switchport mode access
 switchport access vlan 130

! Printer port
interface fa0/11
 switchport mode access
 switchport access vlan 130

! Trunk to router
interface fa0/24
 switchport mode trunk
 switchport trunk allowed vlan 130
```

HO Router (R-HO) – Add Limpopo routes

```
interface g0/2
 ip address 10.22.241.1 255.255.255.252
 no shutdown

! Route to Limpopo LAN
ip route 10.22.32.0 255.255.255.224 10.22.241.2
```

🔹 **Step 5: Ping Tests**

From a Limpopo PC (e.g. 10.22.32.10):

1. ping 10.22.32.1 → test VLAN gateway.
2. ping 10.22.241.2 → test branch router WAN.
3. ping 10.22.241.1 → test HO router WAN.
4. ping 10.22.0.130 → test DNS at HO.
5. ping 10.22.16.5 → test a Mpumalanga PC (branch-to-branch via HO).

✅ If all succeed → Limpopo is fully connected to HO & other branches.

0) Standardise WAN addressing (final)

To avoid confusion, we will use the **10.22.240.0/30 block** split into /30s for each HO↔branch WAN:

- **HO ↔ Mpumalanga (MPU)** → 10.22.240.0/30
  - **R-HO (HO side):** 10.22.240.1
  - **R-MPU (MPU side):** 10.22.240.2
- **HO ↔ Limpopo (LIM)** → 10.22.240.4/30
  - **R-HO:** 10.22.240.5
  - **R-LIM:** 10.22.240.6
- **HO ↔ Eastern Cape (EC)** → 10.22.240.8/30
  - **R-HO:** 10.22.240.9
  - **R-EC:** 10.22.240.10

(If you already configured different WAN IPs on R-LIM or others, paste these new ones — I'll show the updated snippets below.)

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1) Devices & cables (EC)

Devices

- Router: R-EC (Cisco 2811/2911)

- Switch: SW-EC-1 (Cisco 2960, 24-port, PoE optional)
- PCs: EC-PC01 … EC-PC10 (10 PCs)
- Printer: EC-Printer-01
- (Optional) Admin PC

Cables

- **R-EC G0/0 ↔ R-HO G0/X** (WAN) — **Copper straight-through** (both GigabitEthernet) set IPs above.
- **R-EC G0/1 ↔ SW-EC Gi0/1** — **Copper straight-through** (trunk)
- **SW-EC Fa0/1–Fa0/10 ↔ PCs** — **Copper straight-through**
- **SW-EC Fa0/11 ↔ Printer** — **Copper straight-through**

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2) Eastern Cape addressing & VLANs (as agreed earlier)

VLAN Name	VLAN ID	Subnet	Default GW	DHCP Range
Users	310	10.22.48.0 /27	10.22.48.1	10.22.48.2 – 10.22.48.30
Mgmt/Printers	399	10.22.48.32 /27	10.22.48.33	10.22.48.34 – 10.22.48.62

Printer static: 10.22.48.34 (example; reserve in exclusions)

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3) CLI — R-EC (copy & paste into Router R-EC)

```
enable
configure terminal
hostname R-EC
no ip domain-lookup
service timestamps log datetime msec
service timestamps debug datetime msec

! ---------------- WAN to HO ----------------
interface GigabitEthernet0/0
 description WAN-to-HO
```

```
 ip address 10.22.240.10 255.255.255.252
 no shutdown

! ---------------- LAN trunk to local switch ----------------
interface GigabitEthernet0/1
 no ip address
 no shutdown

interface GigabitEthernet0/1.310
 encapsulation dot1Q 310
 ip address 10.22.48.1 255.255.255.224

interface GigabitEthernet0/1.399
 encapsulation dot1Q 399
 ip address 10.22.48.33 255.255.255.224

! ---------------- DHCP pools ----------------
ip dhcp excluded-address 10.22.48.1 10.22.48.5
ip dhcp excluded-address 10.22.48.33 10.22.48.35
ip dhcp excluded-address 10.22.48.34

ip dhcp pool EC-Users
 network 10.22.48.0 255.255.255.224
 default-router 10.22.48.1
 dns-server 10.22.0.130
 lease 2

ip dhcp pool EC-MGMT
```

```
 network 10.22.48.32 255.255.255.224
 default-router 10.22.48.33
 dns-server 10.22.0.130
 lease 2

! ---------------- Default route to HO ----------------
ip route 0.0.0.0 0.0.0.0 10.22.240.9

end
write memory
```

Notes:

- We used dns-server 10.22.0.130 (HQ DNS).
- Excluded addresses reserve gateways/printer static. Adjust printer static if you choose another IP.

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4) CLI — SW-EC-1 (copy & paste into SW-EC-1)

```
enable
configure terminal
hostname SW-EC-1
no ip domain-lookup

! Create VLANs
vlan 310
 name EC_Users
vlan 399
 name EC_MGMT

! Access ports: users
```

```
interface range fastEthernet0/1 - 10
 switchport mode access
 switchport access vlan 310
 spanning-tree portfast

! Printer port
interface fastEthernet0/11
 switchport mode access
 switchport access vlan 399
 spanning-tree portfast

! Trunk to router
interface gigabitEthernet0/1
 switchport mode trunk
 switchport trunk allowed vlan 310,399
 no shutdown

! Optional management SVI
interface vlan 399
 ip address 10.22.48.35 255.255.255.224
 no shutdown

end
write memory
```

Note: on a layer-2 2960 the SVI for management is optional; if used, ensure ip default-gateway 10.22.48.33 is set instead.

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5) HO router changes (R-HO) — add EC WAN and routing back to EC

Paste into R-HO:

```
enable
configure terminal

! Add interface for EC WAN (if HO has spare G0/port)
interface GigabitEthernet0/3
 description WAN-to-EC
 ip address 10.22.240.9 255.255.255.252
 no shutdown

! Add route back to EC LAN
ip route 10.22.48.0 255.255.255.224 10.22.240.10

! Ensure routes for MPU and LIM exist (example):
ip route 10.22.16.0 255.255.255.224 10.22.240.2   ! MPU
ip route 10.22.32.0 255.255.255.224 10.22.240.6   ! LIM

end
write memory
```

If R-HO used different interface numbers earlier, put the EC WAN on whatever physical interface is free and use the matching IP.

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6) Cabling & port mapping recap (exact)

- **R-HO G0/3** ↔ **R-EC G0/0** (straight-through) → WAN 10.22.240.9 ⇄ 10.22.240.10
- **R-EC G0/1** ↔ **SW-EC Gi0/1** (straight-through) → trunk carrying VLANs 310,399
- **SW-EC Fa0/1–Fa0/10** ↔ **EC-PC01–EC-PC10** (straight-through)
- **SW-EC Fa0/11** ↔ **EC-Printer-01** (static)

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7) Verification pings & tests (after you apply configs)

Run these **in order** — run on devices in Packet Tracer exactly as shown.

On SW-EC-1 (optional checks)

show vlan brief

show interfaces trunk

show running-config

On R-EC (router)

show ip interface brief ! ensure g0/0 and subinterfaces are up

show ip route ! confirm default route to HO

show ip dhcp binding ! see DHCP leases

ping 10.22.48.1 ! ping branch gateway (from router)

ping 10.22.240.9 ! ping HO WAN (should succeed)

ping 10.22.0.130 ! ping DNS server at HQ

From an EC-PC (Desktop > Command Prompt)

1. ipconfig → ensure it has an IP in 10.22.48.2–10.22.48.30 and GW 10.22.48.1
2. ping 10.22.48.1 → gateway test
3. ping 10.22.240.10 → local router WAN side (should be reachable from PC)
4. ping 10.22.240.9 → HO WAN IP (test WAN link)
5. ping 10.22.0.130 → DNS server at HO
6. ping <MPU PC IP> e.g., ping 10.22.16.5 → cross-branch test (via HO)

From R-HO (HO router)

ping 10.22.240.10 ! ping EC router (should succeed)

ping 10.22.48.1 ! should succeed if static route back to EC is set

ping 10.22.16.5 ! test reachability to MPU PC

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8) If a ping fails — quick checks

- **PC → Gateway fails:** switch port VLAN wrong or trunk to router not carrying VLAN. show vlan brief on switch.

- **Router subinterface down:** show ip interface brief on branch router — subinterfaces go down if trunk is not set or trunk native mismatch.
- **Router cannot ping HQ WAN IP:** check physical WAN cable between R-EC and R-HO and IP addressing.
- **HO cannot reach branch LAN:** ensure static route on HO points to branch router WAN IP.

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9) Save configs

On each device after tests:

write memory